

ABSTRACT

AN AUTONOMOUS CREDIT RISK ASSESSMENT AND EXPLAINABLE AI SYSTEM WITH HYBRID MULTI-BANK POLICY RAG ENGINE

(Loan-IQ)

Loan-IQ is an intelligent, autonomous credit risk assessment and self-healing underwriting system designed to ensure reliable, transparent, and uninterrupted financial decision-making, especially during complex credit evaluations and multi-bank policy assessments. The system continuously monitors applicant risk profiles in real time, detects underwriting anomalies or policy violations using advanced AI/ML techniques, and automatically takes corrective actions to generate audit-ready risk scoring and decision rationale without manual intervention bottlenecks.

It combines a 2-tier Stacking Ensemble Classifier (XGBoost, LightGBM, CatBoost, Random Forest with an Artificial Neural Network Meta-Learner), SHAP explainable AI attributions, and a specialized multi-bank policy compliance module. When a loan application or policy query is submitted, Loan-IQ evaluates financial obligations (FOIR, LTV, EMI), isolates policy violations, and reroutes non-compliant requests with actionable mitigation steps. In parallel, it utilizes a Hybrid RAG Engine combining dense FAISS vector search and sparse BM25 keyword retrieval with Reciprocal Rank Fusion (RRF) re-ranking to ensure that critical bank guidelines (SBI, HDFC, ICICI, IOB, Canara, RBI) reach decision-makers accurately.

Furthermore, the architecture integrates comprehensive compliance protocols aligned with RBI Master Directions (KYC/AML) and Video Customer Identification Processes (V-CIP). It provides granular local SHAP feature impact attributions that explain the exact mathematical weight of credit bureau scores, FOIR ratios, and income factors behind every decision, fulfilling modern banking explainable AI (XAI) mandates. This dual capability ensures both high predictive accuracy and total regulatory transparency for diverse credit products including Home, Personal, Auto, and Education loans.

Additionally, Loan-IQ is engineered for seamless cloud containerization and distributed API microservice integration across web and mobile platforms. Hosted on high-performance cloud environments with automated health checks, lazy model loading, and lightweight REST/Gradio interfaces, the architecture provides sub-second latency for real-time credit decisioning. This scalable design enables seamless interoperability with core banking systems, loan origination platforms (LOS), and automated credit bureau APIs.

The system uses AI to predict creditworthiness, learn from historical lending data, and improve its response over time. With a user-friendly dashboard, it provides real-time alerts, credit risk visualization, and detailed logs for better decision-making. Loan-IQ aims to create a resilient, secure, and intelligent financial technology infrastructure that supports continuous, compliant credit processing in disaster, high-risk, and enterprise banking scenarios.

In summary, Loan-IQ establishes a next-generation paradigm for digital lending by harmonizing data-driven credit risk classification with explainable AI and automated policy compliance. By eliminating subjective manual underwriting delays and enforcing uniform risk standards across institutional lenders, the system mitigates non-performing asset (NPA) risks while accelerating loan disbursement workflows. Ultimate deployment benchmarks demonstrate that Loan-IQ provides a highly reliable, scalable, and audit-compliant framework for modern financial institutions.

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